1. Our Groups

We are interested in semidirect products of the form $T \rtimes_\varphi S$ where $T = \langle t \rangle$ is a cyclic group of order e, $S = \langle s \rangle$ is a cyclic group of order f, and $\varphi : S \to \mathrm{Aut}(T)$ is defined by $\varphi(s)(t) = t^p$ for some prime p. For this to make sense we need some conditions on e, f, and p. These turn out to be $p \nmid e$ and $p^f \equiv 1 \pmod{e}$.

To see this, notice that the two things we need are that $(t \mapsto t^p)$ is an automorphism of T and that φ is a homomorphism.